BESS Requirements Document



Team Members:

Connor Solberg

Wouter Henderikx

Jelle Reuser

Min Jeong Paek

Date: XX/09/2026

Version: X

Revision Table

Version	Date	Changes
V1	14/10/2026	First Version
V2	14/10/2026	Requirements sorted into Electrical, Mechanical, and Software; feedback table and notes merged into requirements; battery updated to 4S3P
V3	26/10/2026	Fleshed out changes and applied testing standards to each requirement.

Distributor List

Member	V.1	V.2	V.3
TE'GETHER			
Connor Solberg	X	X	X
Min Jeong Paek	X	X	X
Wouter Hendrikx	X	X	X
Jelle Reuser	X	X	X
TE Connectivity			
Rik van Bemmel	X	X	
Avans Hogeschool			
Bob Bus		X	

Abbreviations and Definitions

Definition	
BESS	Battery Energy Storage System
BMS	Battery Management System
AFE	Analog Front End (cell-monitoring IC)
BOM	Bill of Materials
CC/CV	Constant Current / Constant Voltage (charging profile)
E-stop	Emergency stop
EMC	Electromagnetic compatibility
FoS	Factor of Safety
HMI	Human–Machine Interface
IP / IPXXB	Ingress protection code (IEC 60529); IPXXB = protected against access with a finger
LC	Load case (Table F2)
$LiFePO_4$ / LFP	Lithium Iron Phosphate (battery chemistry)
MCU	Microcontroller Unit
MoSCoW	Must / Should / Could / Won't (priority)

NC	Normally Closed (contact)
NTC	Negative Temperature Coefficient thermistor
OV / UV / OC / OT / UT	Overvoltage / Undervoltage / Overcurrent / Over-temperature / Under-temperature
PCB	Printed Circuit Board
PE	Protective earth
PSU	Power supply unit (AC/DC)
PWM	Pulse-Width Modulation
SELV	Safety extra-low voltage (≤ 60 V DC)
SoC / SoH	State of Charge / State of Health
SWD	Serial Wire Debug
T / A / I / D	Verification methods: Test, Analysis, Inspection, Demonstration
TBD	To Be Defined
UART	Universal Asynchronous Receiver-Transmitter

Requirements are sorted into Electrical, Mechanical and Software, followed by project requirements. The column “V1 ref.” shows where each requirement came from in version 1: a requirement number, “FB” for the feedback table, or “Notes” for the draft notes. Values that still need a decision are marked TBD and are collected in the open items at the end. The column “Ver.” gives the verification method: T = Test, A = Analysis, I = Inspection, D = Demonstration. Tables F1 and F2 define the design currents and load cases (LC) that the protection, thermal, wiring and test requirements refer to. Section 6 gives the reasoning behind the key decisions.

Table of Contents

1 Electrical Requirements 6

1.1 System 6

Table D – Battery parameters 6

Table F1 – Design currents (current ladder) 7

Table F2 – Load cases 8

1.2 Measurement 8

1.3 Protection 9

Table E – Protection limits 10

1.4 Cell balancing 11

1.5 Charging 11

1.6 Energy output 12

1.7 Emergency stop and manual disconnect 12

1.8 Power supply, controls and fuses 13

1.9 PCB 14

1.10 Wiring and connectors 15

1.11 Mains (AC) input 15

2 Mechanical Requirements 16

2.1 Enclosure and dimensions 16

Table I – Safety signs and labels 17

2.2 Mounting and layout 17

2.3 Cable routing 18

2.4 Environmental conditions 19

Table G – Component temperature ratings and heat dissipation 19

2.5 Thermal performance 20

Table H – Wire current-carrying capacity (reference) 21

3 Software Requirements 21

3.1 General 21

3.2 Configuration 23

3.3 BMS firmware behaviour 23

3.4 Error codes 25
Table A – Error codes 26
3.5 Communication 26
3.6 Fan control 27
Table C – Fan speed table 28
Table C2 – Fan speed table, enclosure air (NTC3) 29
3.7 PC connection 29
3.8 User interface 29
Table B – Information shown to the user 30
3.9 Graphs / HMI 31
3.10 Data storage 32
3.11 Remote website 33
Live error status 34
Email alerts 34
4 Project Requirements 35
5 Open Items 36
6 Rationale for key decisions 37
6.1 Electrical 38
6.2 Mechanical and thermal 42
6.3 Software 43
6.4 Project 43

1 Electrical Requirements

1.1 System

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E1.1	The BESS shall be able to charge and discharge the 4S3P $LiFePO_4$ battery pack.	–	Must	T	1.1
E1.2	The BESS shall use a 4S3P $LiFePO_4$ battery pack as its energy-storage system, with the parameters in Table D.	12 cells (4S3P) 12.8 V nominal 9.6 Ah	Must	I	1.2, FB
E1.3	The BESS shall monitor the operating condition of the battery during both charging and discharging.	–	Must	T	1.3
E1.4	The BESS shall prevent charging when a battery safety limit is exceeded.	–	Must	T	1.4
E1.5	The BESS shall prevent discharging when a battery safety limit is exceeded.	–	Must	T	1.5
E1.6	The BESS shall provide a means of determining whether the battery is in a charging, discharging or idle state.	–	Should	D	1.6
E1.7	The battery shall normally be operated within the SoC window.	20–85 % SoC	Should	D	Notes
E1.8	The power path (SSRs, fuse holders, wiring, connectors and PCB power traces) shall be rated for the system design current. Upgrading from the present load to the design current shall only need a larger F2 fuse and new protection settings (Table F1).	Design: 10 A continuous, 12 A peak Today: ≈ 5 A Battery: 28.8 A continuous	Must	A	New

Met opmerkingen [CL1]: Verification method: T = Test, A = Analysis, I = Inspection, D = Demonstration. The column itself was added without change tracking; its contents are tracked.

Table D – Battery parameters

Parameter	Per cell	4S3P pack
Number of cells	–	12 (4 series × 3 parallel)
Nominal voltage	3.2 V	12.8 V
Charging voltage	3.65 V	14.6 V
Capacity	3200 mAh	9600 mAh
Maximum charging current	3.2 A	9.6 A
Maximum continuous discharge current	9.6 A	28.8 A
Charge/discharge rate	1C – 2C (charging only from 10 °C at ≤ 0.83C; confirm 1C at 10 °C in the datasheet)	1C = 9.6 A
Shutdown voltage	2.5 V	10 V
Charging temperature	0 to 55 °C	0 to 55 °C
Discharging temperature	−20 to 60 °C	−20 to 60 °C
Storage temperature	< 3 months: −20 to 45 °C; > 3 months: 0 to 25 °C (not in datasheet; team choice)	Same as cell

Met opmerkingen [CL2]: Team decision 20/10: the pack runs at 9.6 A continuous. Table D lists 9.6 A per CELL (= 28.8 A for the pack, i.e. 3C). Check the cell datasheet: if the cell really is 3C, then 9.6 A is a design choice rather than a battery limit, and Table F1 and the report should say so.

Table F1 – Design currents (current ladder)

Each level must sit above the one before it, including tolerances. Everything in the power path is rated for the system design current; the battery is the only part limited below it.

Level	Value	Set by	Purpose
Charger CC setpoint	8.0 A (±5 % → ≤ 8.4 A)	E5.3	Normal charging stays below the charge OC trip
Charge OC trip (AFE)	> 9 A (±5 % → 8.55–9.45 A)	E3.6	Trips before the battery maximum charge current
Battery max. charge current	9.6 A (1C)	Table D	Lowest maximum in the charge path
Battery continuous discharge	28.8 A (3C)	Team decision / Table D	Battery limit, not the system limit
Present load (USB, 45 W total)	≈ 5 A (45 W total USB limit, 90 %, 10 V)	E6.1	Normal operating point today (LC1)
Discharge overload (firmware)	Today: > 5.5 A for > 10 s Upgrade: > 10 A for > 10 s	E3.18	Allows short peaks, stops sustained overload
System peak discharge	12 A for ≤ t_peak = 10 s (upgrade)	System design	Short peaks (LC3)
System design current	10 A continuous	E1.8	Rating of everything except the battery (upgradability)
Discharge OC trip (AFE)	Today: > 6.5 A (±5 % → 6.2–6.8 A) Upgrade: > 13 A (±5 % → 12.35–13.65 A)	E3.6	Fast hardware protection above the design current
SSRs K1, K2	≥ 15 A	E7.6	At least the fuse rating
Fuses F1, F2	F1: 15 A F2: 7.5 A today, 15 A upgrade	E8.5	Continuous current ≤ 75 % of rating; protects the wiring
Power wiring ampacity (derated)	≥ 15 A	E10.1	≥ max(1.5 × 10 A, fuse rating)
Current measurement range	±20 A	E2.3	Covers every trip level

Table F2 – Load cases

Case	Condition	Used for
LC1	Continuous discharge at the present full load (USB outputs at the 45 W total limit, ≈ 5 A) from 85 % to 20 % SoC, charger off, fans on automatic, lab ambient (record the inlet air temperature).	Thermal steady state and temperature rise (M5.1, M5.2); wire and contact temperatures
LC2	Charge at the charger setpoint (8.0 A) CC/CV from 20 % to 85 % SoC, balancing active.	Charger PSU and balancing heat (M5.1, M5.3)
LC3	Discharge at 12 A for t_peak (10 s), then back to the present load. Requires the upgrade settings and F2 = 15 A.	Peak function, overload protection (E3.18), SSR and joint temperatures
LC4	Idle (no battery current), all auxiliaries on (Pi, screen, fan controller, K1/K2 switched on) for at least 4 h.	Heat from auxiliaries only (M5.3)
LC5	Start-up after soaking at the minimum operating temperature (0 °C), then a charge attempt. Test if a cold chamber is available; otherwise simulate with a cooled NTC.	Under-temperature behaviour (E3.4, Table E)
LC6	Upgrade design point: 10 A continuous discharge with 12 A peaks (F2 = 15 A). Analysis; test only after the upgrade.	Sizing of wiring, fuses, SSRs and connectors (E1.8, E10.1)

1.2 Measurement

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E2.1	The BMS shall measure the voltage of each of the 4 series cell groups.	Range: 2.5–3.65 V Resolution: 1 mV Accuracy: ±10 mV at 25 °C Update: ≤ 200 ms	Must	T	2.1, FB
E2.2	The BMS shall measure the total battery-pack voltage.	–	Must	T	2.2
E2.3	The BMS shall measure the total pack current flowing into and out of the battery, including overcurrent.	Range: ±20 A Resolution: 10 mA Update: 100 ms	Must	T	2.3, FB
E2.4	The BMS shall monitor the temperature at multiple points using NTC thermistors: NTC1 on a core cell, NTC2 on the terminal (discharge contact) joint and NTC3 in the airflow directly downstream of the battery pack.	3 sensors	Must	I	2.4, FB
E2.5	The temperature sensors shall be NTC thermistors with the listed properties.	R25: 10 kΩ ±1 % B: 3950 K ±1 % Time constant: ≤ 15 s Dissipation: ≈ 5 mW/°C (still air)	Should	I	FB
E2.6	Temperature shall be measured over the full operating range with the required accuracy.	Range: −20 to 100 °C Accuracy: ±1 °C (target ±0.5 °C)	Must	T	FB
E2.7	Each temperature shall be sampled periodically.	1 sample/s Slowest: 1 sample/min	Must	T	FB

1.3 Protection

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E3.1	The BMS shall detect when an individual cell group exceeds its specified maximum voltage (Table E).	See Table E	Must	T	2.5
E3.2	The BMS shall detect when an individual cell group falls below its specified minimum voltage (Table E).	See Table E	Must	T	2.6
E3.3	The BESS shall protect the battery against overvoltage, undervoltage and excessive temperature using the trip conditions, delays and recovery conditions in Table E.	See Table E	Must	T	6.2, 6.3, 6.5
E3.4	The BMS shall prevent charging when the battery temperature is outside the charging temperature range in Table E.	See Table E	Must	T	2.9
E3.5	The BMS shall prevent discharging when the battery temperature is outside the discharge temperature range in Table E.	See Table E	Must	T	2.10
E3.6	The BMS shall detect overcurrent during charging and discharging using AFE autonomous protection.	Charge: > 9 A Discharge: > 6.5 A today (> 13 A upgrade) Accuracy: ±5 % Detection: 100 ms Disconnect: ≤ 120 ms (Table F1)	Must	T	2.11, 6.4, FB
E3.7	The BMS shall detect a short circuit at the battery output.	Threshold: ≈ 60 A (nearest AFE setting) Detection: ≤ 200 µs	Must	T	2.12, FB
E3.8	The BMS shall automatically disconnect the battery from the system when a critical protection limit is exceeded or a critical fault is detected.	–	Must	T	2.7, 6.1
E3.9	Safety flags shall be OV, UV, OC, OT and UT. Critical faults shall be: short circuit, E-stop, AFE communication loss and invalid safety-sensor data.	–	Must	I	FB
E3.10	Loss of valid AFE or sensor data for longer than the timeout shall be treated as a critical fault.	Timeout: 500 ms	Must	T	FB
E3.11	During a safety lockout, the charge/discharge MOSFETs shall be OFF, balancing shall be OFF, K1 and K2 shall be OFF, a latched fault code shall be stored, and monitoring and communication shall stay active where possible.	–	Must	T	FB

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E3.12	Recovery from a safety lockout shall only be possible when the fault is removed, valid data has been received for the set time and an explicit reset is given. A reset shall be rejected while the fault is still active.	Valid data: 5 s	Must	T	FB
E3.13	The BESS shall prevent the battery from being charged or discharged when the BMS is not operating correctly.	–	Must	T	6.8
E3.14	Each temperature sensor shall have its own protection function; readings shall not be averaged. Where several sensors apply, the highest valid reading shall be used.	–	Must	T	FB
E3.15	NTC1 (core cell) shall be used for battery over- and under-temperature protection (Table E) and shall cut off the battery at the overheating limit (E05).	Cut-off: > 55 °C (Table E)	Must	T	FB
E3.16	NTC2 (terminal joint) shall raise fault E16 (TERMINAL_OVERTEMP) when it exceeds its absolute limit or is much hotter than NTC1.	> 70 °C, or NTC2 – NTC1 > 15 °C	Must	T	FB
E3.17	NTC3 (air downstream of the battery) shall raise warning E17 (ENCLOSURE_OVERTEMP) above its limit, and the fans shall then run at 100 %.	> 45 °C (proposed)	Should	T	New
E3.18	Sustained discharge above the battery continuous rating shall be stopped by the firmware (overload, E04), while short peaks up to the peak current in Table F1 are allowed.	Today: > 5.5 A Upgrade: > 10 A for longer than t_peak = 10 s	Must	T	New

Table E – Protection limits

Protection	Trip condition	Detection delay	Recovery condition	Action
Cell-group OV	Any group ≥ 3.60 V	1 s	All groups ≤ 3.45 V for 5 s	Charge OFF
Cell-group UV	Any group ≤ 2.70 V	1 s	All groups ≥ 2.90 V for 5 s	Discharge OFF
Charge OC	Charge current > 9 A; threshold accuracy ±5 %	100 ms target; disconnect ≤ 120 ms	Cause removed + explicit reset	Charge OFF (AFE autonomous protection)
Discharge OC	Discharge current > 6.5 A today (> 13 A upgrade); threshold accuracy ±5 %	100 ms target; disconnect ≤ 120 ms	Cause removed + explicit reset	Discharge OFF (AFE autonomous protection)
Discharge overload	Discharge current > 5.5 A today (> 10 A upgrade) for longer than t_peak (10 s)	t_peak	Cause removed + explicit reset	Discharge OFF (firmware), E04
Charge UT	Any battery sensor < 5 °C	1 s	All battery sensors ≥ 10 °C for 5 s	Charge OFF
Charge OT	Any battery sensor > 50 °C	1 s	All battery sensors ≤ 45 °C for 5 s	Charge OFF
Discharge UT	Any battery sensor < −10 °C	1 s	All battery sensors ≥ −5 °C for 5 s	Discharge OFF

Protection	Trip condition	Detection delay	Recovery condition	Action
Discharge OT	Any battery sensor > 55 °C	1 s	All battery sensors ≤ 50 °C for 5 s	Discharge OFF
Short circuit	Discharge current > 60 A (nearest AFE setting)	≤ 200 µs target	Cause removed + valid data for 5 s + explicit reset; rejected while fault active	Charge/discharge/balancing OFF + latched fault code; K1/K2 OFF
Terminal OT (NTC2)	NTC2 > 70 °C, or NTC2 − NTC1 > 15 °C	1 s	Cause removed + valid data for 5 s + explicit reset	Charge/discharge OFF, latched E16
Enclosure air OT (NTC3)	NTC3 > 45 °C	1 s	NTC3 ≤ 40 °C for 5 s	Warning E17, fans 100 %

1.4 Cell balancing

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E4.1	The BMS shall provide cell balancing to reduce voltage differences between the cell groups.	–	Must	T	2.8
E4.2	Balancing shall start and stop at set cell-group voltage differences.	Start: ΔV ≥ 30 mV Stop: ΔV ≤ 10 mV	Should	T	FB
E4.3	Passive balancing shall use bleed resistors sized for the balancing current below.	Balancing current: 50–100 mA Resistor: 33–68 Ω Heat: ≤ 0.33 W per channel	Should	A	FB

Met opmerkingen [CL3]: Heat: at 3.3 V a 15 Ω bleed resistor dissipates about 0.73 W per channel (about 2.9 W if all 4 channels balance at once). Included in Table G and the heat budget.

1.5 Charging

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E5.1	The BESS shall use an internal AC/DC charger (see 1.11) suitable for charging a 4S $LiFePO_4$ battery pack.	14.6 V charge voltage	Must	I	3.1
E5.2	The charger shall use a charging profile suitable for $LiFePO_4$ batteries (CC/CV).	–	Must	T	3.2
E5.3	The charger constant-current setpoint shall stay below the charge overcurrent trip (E3.6), including both tolerances, so that normal charging never trips the protection (Table F1).	8.0 A ±5 % (0.83C)	Must	T	3.3
E5.4	The charger shall limit the charging voltage to the maximum charging voltage of the battery (Table D).	14.6 V (3.65 V/cell)	Must	T	3.4
E5.5	The BMS shall be able to enable and disable the charger.	–	Must	T	3.5
E5.6	The BMS shall disable charging when a protection limit in Table E is exceeded.	See Table E	Must	T	3.6
E5.7	The BESS shall indicate when the battery is being charged.	–	Should	D	3.7

Met opmerkingen [CL4]: Tolerance stack: the battery max of 9.6 A is the lowest maximum in the charge path. Charge OC trip 9 A ±5 % = 8.55–9.45 A stays below 9.6 A. Charger 8.0 A ±5 % = max 8.4 A stays below 8.55 A, so normal charging never trips the protection. Check the real charger tolerance in its datasheet.

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E5.8	The system shall be considered CHARGING when charging is enabled and the current passes the start threshold, and shall leave CHARGING at the stop threshold or when charging is disabled. The state shall be shown as a CHARGING status code on the Pi HMI. CHARGING does not mean the battery is full.	Start: ≤ −0.20 A for 1 s Stop: ≥ −0.10 A for 1 s Update: 1 s	Should	T	FB
E5.9	A charging curve shall be developed for the selected cells.	–	Could	A	Notes

1.6 Energy output

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E6.1	The BESS shall supply energy to external loads through 1 × USB-A and 2 × USB-C outputs. The USB-C outputs shall support USB Power Delivery (PD) fast charging; the USB-A output gives 5 V. The total output of all ports shall be limited (power sharing) so the battery current stays within Table F1. Each output shall have overcurrent, short-circuit and overvoltage protection.	USB-A: 5 V, 3 A (15 W) USB-C: PD, 5–20 V Total: ≤ 45 W today; ≤ 95 W after upgrade	Must	T	4.1, FB
E6.2	The BESS shall disconnect the load when a battery protection limit is exceeded.	–	Must	T	4.2
E6.3	The BESS shall limit the maximum output current to the system output limits in Table F1.	Today: ≈ 5 A Upgrade: 10 A continuous, 12 A for ≤ 10 s	Must	T	4.3
E6.4	The BESS shall indicate when energy is being supplied to an external load.	–	Should	D	4.4
E6.5	The output shall be switchable on and off with a rocker switch.	–	Should	D	FB

1.7 Emergency stop and manual disconnect

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E7.1	The BESS shall have a manually accessible method of disconnecting the battery from the rest of the system.	–	Must	D	6.7
E7.2	The E-stop shall be a latching device with normally closed (NC) channels, wired to a safety relay that removes the drive of the charge switch K1 and the discharge switch K2 (DC solid-state relays) (Figure 1).	2 NC channels	Must	I	FB
E7.3	The E-stop shall cut power in hardware and shall not depend on the MCU firmware.	–	Must	T	FB
E7.4	The time from pressing the E-stop to K1 and K2 opening shall stay below the reaction-time limit.	< 100 ms	Must	T	FB

Met opmerkingen [CL5]: If K1/K2 become solid-state relays or MOSFETs instead of contactors: replace 'coil drive' with 'drive', and note that E7.5 relies on K1/K2 feedback contacts, which SSRs/MOSFETs do not have. Another feedback method (e.g. voltage sensing after the switch) would then be needed.

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E7.5	After an E-stop, the system shall only restart when the E-stop is released and the RESET button on the safety relay is pressed. Before the reset is accepted, K1 and K2 shall be confirmed off by sensing the voltage on the output side of each switch (SSRs have no feedback contacts).	–	Must	T	FB
E7.6	K1 and K2 shall be rated above the maximum system current.	≥ 15 A (≥ fuse rating F1/F2)	Must	I	FB
E7.7	The fans shall be de-energised by the E-stop (fan supply switched through the safety relay). The E-stop overrides every fan command, including 100 % during E05 and the hardware fallback (S6.7). Fan openings shall be guarded (M1.3).	Fans OFF	Must	T	FB

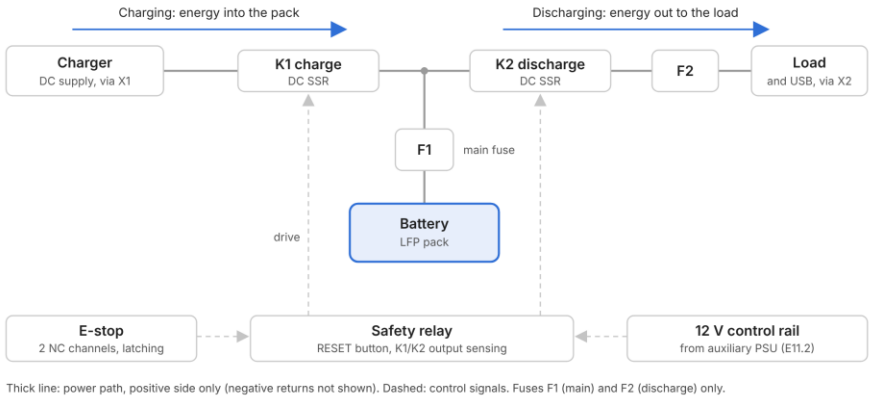


Figure 1 – E-stop and safety relay circuit

1.8 Power supply, controls and fuses

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E8.1	The BMS shall be powered from a dedicated 5 V input. Its power consumption shall be defined after component selection.	5 V input Range: 4.75–5.25 V (±5 %) Power: after component selection	Must	T	FB
E8.2	When any cell group falls below the shutdown voltage, discharging shall be disabled. Charging shall remain possible if there are no other faults (temperature, cell voltage).	< 2.5 V per group (pack 10 V)	Must	T	FB
E8.3	The BMS shall be switchable off with a hardware button (potentially switching the entire control rail).	12 V rail	Should	D	FB

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E8.4	The BESS shall have the following physical controls. Operator controls, usable with the door closed: E-stop, On/Off for the BMS, On/Off for the output and the safety-relay RESET. Service controls, inside the cabinet or recessed so they cannot be pressed by accident: Boot, Upload/Programming and Service.	–	Must	I	FB, Notes
E8.5	The BESS shall have a main fuse (F1) as close as possible to the battery positive terminal and a discharge fuse (F2). The charge path is protected by F1, so its wiring shall be rated for F1. Each DC fuse shall have a DC voltage rating ≥ 32 V DC and a breaking capacity above the prospective battery short-circuit current, and shall carry its continuous current at ≤ 75 % of its temperature-derated rating. Fuse type (fast/time-delay) follows from the fuse benchmark.	2 DC fuses F1: 15 A F2: 7.5 A (upgrade: 15 A) Breaking capacity: ≥ 1000 A	Must	I	FB

1.9 PCB

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E9.1	The PCB shall not exceed the maximum outline and board thickness.	≤ 150 × 150 mm 1.6 mm thick Component height: ≤ 40 mm (cabinet space not limiting; confirm in CAD)	Must	I	FB
E9.2	The PCB shall have the specified layer stack-up.	4 layers 1 oz copper	Should	I	FB
E9.3	The PCB material shall be FR-4.	–	Should	I	FB
E9.4	The PCB shall be designed in Altium Designer.	–	Must	I	FB
E9.5	The PCB revision shall be printed on the silkscreen in a fixed format.	R0x	Should	I	FB
E9.6	The PCB shall support testing with test pads, a programming/debug connector, measurement points and jumpers/isolation points.	–	Should	I	FB

1.10 Wiring and connectors

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E10.1	Power wiring shall have a current-carrying capacity, after derating for the internal cabinet temperature and grouping (Table H), of at least the larger of 1.5 × the system design current and the rating of the fuse that protects it. Wire: 2.5 mm^2 (≈ 14 AWG) stranded, tinned copper, insulation rated ≥ 105 °C.	≥ 15 A after derating (1.5 × 10 A = 15 A; F1/F2 max. 15 A)	Must	A	FB, Notes
E10.2	Data wiring shall be routed separately from power and mains wiring, and shall cross them at 90° where crossing cannot be avoided. NTC and I^2C/UART lines shall be twisted pairs (signal with its return). Data wires: 0.25–0.5 mm^2 (AWG 24–20) twisted pairs with locking connectors; shielding only if E10.5 fails.	Separate wire ducts for data and power Crossings at 90°	Should	I	FB
E10.3	Cable insulation shall be selected on temperature rating, flexibility and flame behaviour. Insulation shall be flame retardant (IEC 60332-1 or UL VW-1).	Power: ≥ 105 °C Signal: ≥ 80 °C	Should	I	FB
E10.4	Power connectors shall be specified with connector type, voltage/current rating and polarity keying. Battery and charger connectors shall be keyed so they cannot be mated with reverse polarity or with each other. Contacts shall be rated for the system design current at the internal cabinet temperature.	≥ 12 A per contact (derated)	Must	I, D	FB
E10.5	Temperature readings shall not change by more than the limit when the battery current steps between 0 A and 9.6 A, when K1/K2 switch, and while the fans run at any PWM duty (noise pick-up check).	≤ 0.2 °C	Should	T	New
E10.6	During a 1 h run of LC1 with the fans at 100 % and the charger PSU switched on, the AFE (I^2C) and UART links shall stay below the allowed number of failed messages (bad checksum or no reply).	≤ 1 per 10 000 frames	Should	T	New
E10.7	The MCU and the Raspberry Pi shall not reset or lose communication during repeated E-stop / K1/K2 switching under load.	50 cycles, 0 resets	Should	T	New

1.11 Mains (AC) input

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E11.1	The BESS shall be supplied from the mains through a fixed 3-core mains cord (L, N, PE) that enters the cabinet through a cable gland with strain relief. Inside the cabinet the cord shall end on DIN-rail terminals, followed by a double-pole mains switch and the inlet fuse F0 (E11.6).	230 V AC ±10 %, 50 Hz Cord: H05VV-F 3G1.0 mm^2	Must	I	New

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
E11.2	The AC supply shall feed two AC/DC power supplies inside the cabinet: a charger PSU (battery charging) and an auxiliary PSU (BMS, Pi, screen, fans, safety relay). Both shall be CE-marked with a safety approval (IEC 62368-1 or equivalent).	2 PSUs	Must	I	New
E11.3	All accessible metal parts (cabinet, door, mounting plate) shall be bonded to protective earth (PE); the door through a flexible earth strap.	PE resistance ≤ 0.1 Ω	Must	T	New
E11.4	Mains (AC) wiring shall run in its own wire duct, separate from DC/SELV wiring, and all mains terminals shall be finger-safe (IPXXB) with the door open.	Separate duct; IPXXB	Must	I	New
E11.5	Mains wiring shall use the standard colours (L brown, N blue, PE green/yellow) and a cross-section suited to the inlet fuse.	≥ 1.0 mm^2 (proposed)	Must	I	New
E11.6	The AC side shall be protected by an inlet fuse (F0) and, where required, a fuse per PSU (F4 charger PSU, F5 auxiliary PSU). AC fuses shall be rated ≥ 250 V AC and time-lag (T) for PSU inrush; ratings follow from the fuse benchmark.	F0: T2A (time-lag, 250 V AC) F4/F5: only if a PSU has no internal fuse	Must	I	New
E11.7	All DC circuits inside the cabinet shall be SELV.	≤ 60 V DC	Must	A	New

Met opmerkingen [CL6]: Assumes the existing cabinet is metal (photo in the previous team's System breakdown). If it is plastic, use double/reinforced insulation for the mains parts instead of PE bonding of the cabinet.

2 Mechanical Requirements

2.1 Enclosure and dimensions

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
M1.1	A CAD model with engineering drawings shall be made of the BESS.	–	Must	I	Notes
M1.2	All BESS components, wiring and the battery pack shall fit inside the existing wall cabinet with the door closed, with free space kept in front of the fan inlets and outlets.	Cabinet inner H × W × D: TBD mm (measure for the CAD model; space is not limiting) Free space at fans: ≥ 50 mm (proposed)	Must	I	FB
M1.3	The cabinet shall have IP20 with the door closed, including at the fan openings (fan guards). With the door open, live parts of the battery, DC power path and mains wiring shall be finger-safe (IPXXB).	IP20 (door closed) IPXXB (door open)	Must	T	FB
M1.4	The cabinet shall carry the safety signs and labels listed in Table I.	See Table I	Should	I	FB

Table I – Safety signs and labels

Sign / label	Standard	Location
W012 Electricity	ISO 7010	Door (outside); mains inlet; covers of mains terminals and PSUs
W026 Battery charging	ISO 7010	Door (outside); next to the battery pack
W018 Automatic start-up (fans start by themselves)	ISO 7010	Next to each fan
W017 Hot surface	ISO 7010	Only where the thermal test (M5.3) shows hot accessible surfaces
M002 Refer to instruction manual	ISO 7010	Door (outside)
M021 Disconnect before carrying out maintenance or repair	ISO 7010	Inside of the door
Yellow background behind the red E-stop actuator	ISO 13850	E-stop on the left side wall
Battery rating: $LiFePO_4$, 4S3P, 12.8 V, 9.6 Ah, 123 Wh, max. charge 14.6 V	–	On the battery pack and inside the door
Separate-collection symbol (crossed-out wheeled bin)	EU Batteries Regulation 2023/1542	On the battery pack
Mains rating: 230 V AC, 50 Hz, A1	–	Next to the mains cable entry
Protective earth symbol	IEC 60417-5019	At the PE terminal and PE stud
Control markings: I / O symbols and text for Output On/Off and RESET	IEC 60417-5007 / 5008	Next to each control
Fuse labels: ID and rating (e.g. “F1 20 A”)	–	Next to each fuse holder
Wire and terminal markers matching the wiring diagram	–	Both ends of every wire (M2.6)

2.2 Mounting and layout

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
M2.1	The electronics shall be mounted on the cabinet mounting plate using TS35 DIN rails, terminal blocks and slotted wire ducts, arranged in three zones: mains (AC), DC power and control/signal.	TS35 DIN rail 3 zones	Must	I	FB
M2.2	The E-stop shall be mounted on the left side wall of the cabinet near the battery, not on the door, so it is visible and reachable with the door open or closed. The actuator shall be red on a yellow background (ISO 13850).	Height: ≥ 0.6 m above the floor (EN 60204-1) Final height set in the CAD layout once the cabinet and stands are measured	Must	I	FB
M2.3	NTC1 shall be a bead NTC fixed to the surface of a cell in the middle of the pack. NTC2 shall be a probe NTC fixed to the terminal joint. NTC3 shall be an air-temperature sensor in the airflow directly downstream of the battery pack, before the electronics and the exhaust. NTC4 (if fitted, E2.8) shall be at the air inlet.	–	Should	I	FB

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
M2.4	The cooling airflow shall pass the battery pack before the power electronics and PSUs (inlet → battery → electronics → exhaust). Thermal performance is specified in 2.5.	–	Should	I	Notes
M2.5	Every component shall be replaceable from the front with the door open, using standard hand tools, without removing other components.	–	Should	D	New
M2.6	Every wire shall carry a marker with its wire number at both ends, and every component and terminal shall carry its designation from the wiring diagram (e.g. K1, F2, X1).	–	Should	I	New
M2.7	All components, including the battery pack, and all wiring shall be mechanically fixed (DIN clips, screws, brackets, cable ties on tie mounts). Nothing shall rest loose or hang on its wires.	–	Must	I	New
M2.8	Cables to the door (screen, door controls) shall have a hinge loop with strain relief on both sides and shall meet the flexing bend radius (M3.1).	–	Should	I	New
M2.9	Service connectors (Pi SD card/USB, SWD and service UART headers) shall be reachable with the door open without disassembly.	–	Should	D	New

2.3 Cable routing

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
M3.1	Cables shall not be bent below the minimum bending radius from the datasheet of the selected cable. Fixed wiring and flexing wiring (door hinge) shall each have a defined minimum radius.	Fixed: ≥ 4 × OD Flexing: ≥ 10 × OD (confirm with the wire datasheet)	Should	I	FB
M3.2	Wire ducts shall not be filled beyond the maximum fill of their cross-sectional area.	≤ 50 %	Should	I	Research

2.4 Environmental conditions

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
M4.1	The BESS shall work in an indoor room between 0 °C and 35 °C, and shall only charge when the room is between 10 °C and 35 °C. Why 35 °C: the battery stops charging at 50 °C, the cabinet makes the battery up to 10 °C warmer than the room (checked by M5.1), and 5 °C is kept as margin (50 − 10 − 5 = 35). Why 10 °C: below 10 °C charging is blocked by the protection (Table E).	Room: 0 to 35 °C Charging: 10 to 35 °C	Must	A	FB
M4.2	The BESS shall be stored (unpowered) within the storage temperature range of its most limiting component (Table G).	Battery < 3 months: −20 to 45 °C Battery > 3 months: 0 to 25 °C (not in datasheet; team choice) System: same as battery (check Table G parts)	Should	A	FB
M4.3	The BESS shall operate within the relative humidity range, non-condensing (IEC 61010-1 normal environmental conditions).	Max. 80 % RH up to 31 °C, decreasing linearly to 50 % RH at 40 °C	Could	A	FB
M4.4	The BESS shall operate up to the maximum altitude (default of IEC 61010-1 and IEC 62368-1).	≤ 2000 m	Could	A	FB
M4.5	Before storage or transport, the battery shall be brought to the storage SoC. The SoC check/recharge interval during storage shall be written in the SOP (P1.2).	Storage SoC: 50 % Check and recharge to 50 %: every 3 months	Should	I	New

Table G – Component temperature ratings and heat dissipation

Fill in from datasheets as components are chosen. The lowest maximum rating, minus the internal temperature rise, sets the maximum operating ambient (M4.1). The heat column is the heat budget for the cabinet (W = watts dissipated inside the cabinet).

Component	Rated temp. min / max [°C]	Heat [W] LC1 / LC2 / LC4	Source
Battery cells (4S3P)	Charge 0 / 55; discharge −20 / 60	≈ 2–3 / ≈ 1.5–2 / 0 (I^2R, pack R ≈ 20–27 mΩ, to confirm)	Table D; cell datasheet
Balancing resistors (4×)	TBD	≤ 0.33 per channel (33 Ω at 3.3 V); ≤ 1.3 all four	E4.3
BMS PCB (AFE, MCU, MOSFETs, shunt)	TBD	TBD	Datasheets / measure
Charger PSU (AC/DC)	TBD	0 / ≈ 13 (117 W out at η ≈ 90 %, assumed) / standby TBD	Datasheet
Auxiliary PSU (AC/DC)	TBD	TBD (≈ 10–15 % of its output)	Datasheet
Raspberry Pi	0 / 50 (ambient)	TBD (measure)	Pi product brief
Touchscreen	TBD	TBD	Datasheet
Fan controller module	TBD	TBD	Datasheet
SSRs K1, K2	TBD	I^2 × R_on (SSR datasheet)	Datasheet
Safety relay	TBD	TBD	Datasheet
Fans (2×)	TBD	TBD	Datasheet
NTC sensors	TBD	–	Datasheet
Wiring insulation	≥ 105 (power), ≥ 80 (signal)	I^2R per metre (Table H)	E10.3
Fuses and fuse holders	TBD (temperature derating)	TBD	Datasheet
Connectors / terminal blocks	TBD	TBD	Datasheet
3D-printed parts (if any)	PLA softens ≈ 55–60; PETG ≈ 80; ASA/ABS ≈ 100	–	Material datasheet
Total heat in cabinet	–	LC1 TBD / LC2 TBD / LC4 TBD	Sum of the above

2.5 Thermal performance

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
M5.1	Under LC1 and LC2 (Table F2), the temperature rise of NTC1 above the inlet air shall stay within the limit at steady state or at the end of the test.	≤ 10 K	Must	T	New
M5.2	Under LC1, the largest temperature difference between any two cells shall stay within the limit (temporary thermocouples on at least 4 cells, including the hottest).	≤ 5 °C	Should	T	New
M5.3	Under LC2 and LC4, every component in Table G shall stay below its rated maximum temperature minus the margin, extrapolated to the maximum operating ambient (measured value + 35 °C − lab ambient). The Raspberry Pi shall not throttle.	Margin ≥ 5 K	Must	T	New
M5.4	With one fan disconnected under LC1, warning E12 shall be raised and the battery shall stay within its Table E limits, or the protection shall act.	–	Should	T	New
M5.5	The fans shall run at 100 % when the PWM control signal is lost (4-wire PWM fans).	–	Must	T	New

Met opmerkingen [CL7]: The extrapolation assumes the temperature rise above ambient does not depend on the ambient temperature (reasonable for forced air). This lets us test at room temperature without a climate chamber.

Table H – Wire current-carrying capacity (reference)

Copper conductors, 2 loaded conductors (+ and − of one DC circuit), 30 °C ambient, based on IEC 60364-5-52. Ampacity = table value × ambient factor × grouping factor. Method B = in a wire duct or conduit; Method C = clipped to a surface. Check the values against the standard and the wire datasheet before the final choice.

Cross-section [mm²]	PVC 70 °C, Method B [A]	PVC 70 °C, Method C [A]	XLPE 90 °C, Method B [A]	Resistance, flexible Cu, 20 °C [mΩ/m]
1.0 (≈ 18 AWG)	13.5	15.5	–	19.5
1.5 (≈ 16 AWG)	17.5	19.5	23	13.3
2.5 (≈ 14 AWG)	24	27	31	7.98
4 (≈ 12 AWG)	32	36	42	4.95
6 (≈ 10 AWG)	41	46	54	3.30

Ambient factor	30 °C	35 °C	40 °C	45 °C	50 °C	55 °C
PVC (70 °C)	1.00	0.94	0.87	0.79	0.71	0.61
XLPE / 90 °C	1.00	0.96	0.91	0.87	0.82	0.76

Grouping factor (circuits)	1	2	3	4	6	9
Bunched / in one duct	1.00	0.80	0.70	0.65	0.57	0.50

Example: 2.5 mm² PVC in a shared duct with 3 circuits at 45 °C inside the cabinet: 24 × 0.79 × 0.70 = 13.3 A < 20 A, not enough. 2.5 mm² 90 °C wire routed on its own: 31 × 0.87 × 1.00 = 27 A ≥ 20 A, OK. How the wire is routed changes the wire size.

3 Software Requirements

3.1 General

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S1.1	The BMS firmware on the MCU shall be written in C.	C11	Must	I	8.1
S1.2	The monitoring application on the Raspberry Pi (communication, database, dashboard) shall be written in Rust, with the toolchain version pinned in rust-toolchain.toml.	–	Must	I	8.2
S1.3	The repository shall be organised in clear parts: /firmware (C, MCU), /monitor (Rust, Pi), /protocol (UART message specification) and /docs.	–	Must	I	8.3

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S1.4	The firmware shall be split into modules with one .c/.h pair each: AFE driver, measurement, protection, state machine, SoC/SoH, fan control, LED, UART communication and configuration. Only driver modules shall access hardware registers.	–	Must	I	8.4
S1.5	Logic that does not touch hardware (state machine, SoC calculation, fan curve, limit checks) shall be written so it compiles and runs on a PC for unit testing.	–	Should	T	8.5
S1.6	The firmware shall not use dynamic memory allocation (malloc/free) or recursion. All buffers shall have a fixed size known at compile time.	–	Must	I	8.6
S1.7	The firmware shall use fixed-width integer types (stdint.h) for all measured values and protocol fields (e.g. cell voltage as uint16_t in mV, current as int32_t in mA).	–	Must	I	8.7
S1.8	Firmware and Pi application shall use semantic versioning (MAJOR.MINOR.PATCH). Both versions shall be shown in the dashboard and stored with every logged fault.	–	Should	D	8.15
S1.9	Firmware shall compile with -Wall -Wextra and zero warnings, be formatted with clang-format and pass cppcheck static analysis. Rust code shall pass cargo fmt and cargo clippy with zero warnings.	–	Should	T	8.16
S1.10	The firmware shall have zero MISRA C:2012 mandatory or required violations reported by the cppcheck MISRA add-on, except deviations listed with a rationale in /docs/misra-deviations.md.	–	Could	I	8.17
S1.11	All code shall be kept in one Git repository. Changes shall be merged into main through a pull request reviewed by at least one other team member.	–	Should	I	8.18
S1.12	Unit tests shall exist for every error code (trigger and recovery) and for the fan curve: C firmware tested on a PC with Unity/Ceedling, Rust application with cargo test. All tests shall run automatically on every push (CI).	–	Should	T	8.19
S1.13	Software documentation shall be stored in the repository (/docs) and contain: architecture block diagram, data-flow diagram, state diagram, UART protocol specification, error code table, configuration parameter table, database schema and a dashboard user guide.	–	Must	I	8.20
S1.14	Firmware and Pi application shall each have a README with build, flash and run instructions. Public C functions shall have Doxygen comments; public Rust items shall have rustdoc comments.	–	Should	I	8.21

Met opmerkingen [CL8]: Fan control runs on the Pi (S6.1). Either remove 'fan control' from the MCU module list here and the fan-curve test from S1.12, or define the MCU's role (e.g. only passing on thermostat/tach status).

Met opmerkingen [SR9]: 'Where practical' can never fail a test. Now checked automatically in CI.

3.2 Configuration

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S2.1	All battery parameters shall be defined in one configuration file (bess_config.h): cell count, rated capacity CAP_AH [Ah], cell voltage limits [V], temperature limits [°C], SoC window [%] and fan table. The fan table shall also be sent to the Pi as part of the configuration message (S2.4).	–	Must	I	8.8
S2.2	All current limits shall be defined as C-rates and calculated from CAP_AH (e.g. I_CHG_MAX = 0.83 × CAP_AH). Changing the battery size shall only require changing CAP_AH. A CI check shall fail on any ampere constant defined outside bess_config.h.	–	Must	I	8.9, Notes
S2.3	At start-up the firmware shall check every configuration parameter against the minimum and maximum in the configuration parameter table (S1.13) and check that every upper limit is above its lower limit. If any check fails, the firmware shall keep all outputs off and report E15.	Check done ≤ 500 ms (TBC)	Must	T	8.10
S2.4	The Pi shall receive its battery configuration (including the fan table) from the MCU at start-up, so the Pi application uses the same values without keeping its own copy. The Pi shall not show live values until a valid configuration message has been received.	–	Must	T	8.11

Met opmerkingen [SR10]: Fan table is defined on the MCU but fan control runs on the Pi (S6.1); this states how the Pi gets it.

Met opmerkingen [SR11]: Old example (0.5 × CAP_AH = 4.8 A) contradicted Table E (9 A). Added an automatic check so it is testable.

Met opmerkingen [SR12]: 'Plausible range' was undefined; ranges now live in the configuration parameter table.

Met opmerkingen [SR13]: Check your edit: the Pi now reads the config from a 'setup document', but S2.1 says the config lives in bess_config.h on the MCU, and this sentence still talks about a 'configuration message' and 'no own copy'. Pick one source.

Met opmerkingen [SR14]: Covers the case where the Pi starts or restarts after the MCU.

3.3 BMS firmware behaviour

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S3.1	The MCU shall read cell-group voltages, current and temperatures from the AFE.	Every 100 ms ± 10 ms Voltage ± 10 mV (E2.1) Current: ±20 A range, 10 mA resolution (E2.3); accuracy ± 1 % (TBC) Temp. ± 1 °C (E2.6)	Must	T	Notes
S3.2	When a trip condition in Table E is present for its detection delay, the MCU shall stop charging or discharging as given in the Action column of Table E.	See Table E Delay tolerance ± 100 ms (TBC)	Must	T	Notes
S3.3	When the E-stop feedback input opens, the MCU shall turn all outputs off and report E08 within the set time. (The hardware cut-off itself is covered by E7.3/E7.4.)	≤ 50 ms (TBC)	Must	T	Notes

Met opmerkingen [SR15]: Added rate (from S5.7) and accuracy so the measurement can be checked against a reference meter.

Met opmerkingen [SR16]: Test: one trip + one no-trip (just below threshold) case per Table E row.

Met opmerkingen [SR17]: No reaction time was given for the MCU side.

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S3.4	The MCU shall energise the charge switch K1 only while charging is allowed, and the discharge switch K2 only while discharging is allowed (no active fault blocking that path).	–	Must	T	Notes
S3.5	From power-on until the configuration check (S2.3) has passed and the set number of consecutive valid AFE readings has been received, and during every critical fault, the MCU shall keep all outputs off.	3 valid readings (TBC)	Must	T	Notes
S3.6	The MCU shall only restart charging or discharging when the recovery conditions in Table E are met. The explicit reset shall be given with the reset button on the BMS or the PIN-protected fault reset on the dashboard (S9.12). A reset received while the fault is still active shall be rejected and logged as an event.	See Table E Reset source (TBC)	Must	T	Notes
S3.7	The MCU shall show the system status and active faults with LEDs. Error code n shall be shown as n short blinks followed by a pause. When several faults are active, the LED shall show the fault with the highest severity, and of those the lowest code.	250 ms on / 250 ms off Pause: 2 s (TBC)	Should	D	Notes, 8.14
S3.8	If no valid frame is received from the Pi for longer than the timeout, the MCU shall report warning E11 and, while E11 is active, stop charging (fan control on the Pi is then no longer guaranteed). Discharging and all protection functions on the MCU shall stay active.	Timeout: 5 s (TBC)	Should	T	6.6
S3.9	If a temperature reading is below the open-circuit limit or above the short-circuit limit, the MCU shall reject it and report a sensor fault (E09) within the set time.	Open: < −30 °C Short: > 110 °C Report ≤ 1 s (TBC)	Must	T	FB
S3.10	If a temperature reading differs from the previous valid reading by more than the slew-rate limit of that sensor, the MCU shall discard it.	NTC1: 1 °C/s NTC2: 3 °C/s NTC3/NTC4: 2 °C/s (confirm by test)	Should	T	FB
S3.11	The MCU shall filter each temperature reading with a sliding-window median filter of the set length.	5 samples (TBC)	Should	T	FB
S3.12	If the raw ADC value of a temperature sensor stays exactly the same for longer than the stuck time, the MCU shall flag the sensor as stuck and report E09 (a healthy sensor always shows a few counts of noise).	Raw ADC unchanged: 10 min (confirm by idle logging)	Should	T	FB
S3.13	Each rejected temperature reading shall be logged as an event. If more than n readings of one sensor are rejected within the set time, the MCU shall report the sensor as faulty (E09).	n = 5 (TBC) Window: 60 s (TBC)	Should	T	FB

Met opmerkingen [SR18]: 'Control the relay driver' had no pass/fail condition.

Met opmerkingen [SR19]: End of 'start-up' was undefined.

Met opmerkingen [SR20]: Who/what gives the 'explicit reset' was not defined anywhere. Proposal – please confirm.

Met opmerkingen [SR21]: Blink code was not specified.

Met opmerkingen [SR22]: Check: V.2_claude adds a Pi-independent fan fallback (S6.7: PWM-loss → 100 %, thermostat). With that, stopping charging on Pi loss may no longer be needed. Team decision; value unchanged.

Met opmerkingen [SR23]: DECISION NEEDED: 'safe state' was undefined and E11 is only a Warning. Because the fans run on the Pi (S6.1), this proposal stops charging on Pi loss. Alternative: MCU drives fans to 100 %.

Met opmerkingen [SR24]: Aligned with V.2_claude: measuring range is now −20 to 100 °C (E2.6). Fault limits set 10 K outside that range; −40/125 °C would accept readings outside the specified range as valid. Check against the NTC/ADC curve.

Met opmerkingen [SR25]: 'Outside the NTC range' was not quantified. Check limits against the NTC datasheet.

Met opmerkingen [CL26]: Proposed starting values. Confirm by test: log the largest change per 1 s sample for each sensor during LC1–LC4 and fan on/off steps, then set the limit at about 5x that. Repeated rejections must lead to E09 (S3.13), so a real thermal event is never hidden.

Met opmerkingen [SR27]: Aligned with V.2_claude: per-sensor slew limits (derived from P/(m·c_p) and sensor time constant). Old 2 °C per 100 ms = 20 °C/s was far too loose.

Met opmerkingen [SR28]: S3.10–S3.13: placeholder values added so tests can be written now; tune after the planned sensor tests.

Met opmerkingen [SR29]: Aligned with V.2_claude: current condition removed: ADC noise alone shows a live sensor, so the check also works at idle.

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S3.14	The MCU shall use an independent watchdog. After a watchdog reset the MCU shall start with all outputs off and report E14.	Timeout: 500 ms (TBC)	Must	T	Review
S3.15	The MCU shall estimate SoC by coulomb counting, corrected with the open-circuit voltage after a rest period, within the set accuracy over a full charge/discharge cycle.	± 5 % SoC (TBC)	Must	T	Review
S3.16	After a reset, the MCU shall restore the SoC (from stored value or open-circuit voltage) within the set time.	≤ 2 s (TBC)	Should	T	Review
S3.17	While charging, when the cell-group voltage difference reaches the start value, the MCU shall enable balancing on the highest groups until the difference is at or below the stop value. Balancing shall be off during any fault.	Start: ΔV ≥ 30 mV Stop: ΔV ≤ 10 mV	Must	T	E4.1, E4.2
S3.18	If the cell-group voltage difference stays above the imbalance limit for the set time, the MCU shall report warning E13.	ΔV > 50 mV (TBC) Time: 10 s (TBC)	Should	T	Review
S3.19	The MCU shall disable the charger whenever the charge path is blocked by a fault.	–	Must	T	E5.5
S3.20	When the AFE reports an overcurrent or short-circuit flag, the MCU shall report E03, E04 or E07 within the set time.	≤ 100 ms (TBC)	Must	T	E3.6, E3.7
S3.21	If no valid AFE reading is received for longer than the timeout, the MCU shall report E10 and turn all outputs off.	500 ms (E3.10)	Must	T	E3.10
S3.22	All active faults shall be reported. Where only one can be shown (LED, banner), the priority shall be Critical > Protection > Warning, then lowest code first.	–	Should	D	Review
S3.23	If the discharge current stays above the continuous limit for longer than t_peak, the MCU shall open the discharge path and report E04 (latched; explicit reset). Shorter peaks up to the peak current (Table F1) shall be allowed.	Today: > 5.5 A Upgrade: > 1.04 C (= 10 A) t_peak: 10 s	Must	T	E3.18

Met opmerkingen [SR30]: S3.14–S3.22 are new: these behaviours were referenced (E14, E13, E10, SoC, balancing, charger enable) but had no software requirement to test.

3.4 Error codes

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S4.1	Every fault shall have a unique error code (Exx), name, severity and plain-language description as listed in Table A. The same code shall be used in the firmware, UART messages, LED blink code, dashboard and database. The firmware, Rust application, dashboard and documentation error tables shall be generated from one source file, and a CI check shall confirm they are identical.	–	Must	I	8.14

Met opmerkingen [SR31]: Makes 'the same code everywhere' testable automatically. Also see new codes E16–E18.

Severity: Warning = logged and shown only; Protection = charging or discharging stopped, recovers automatically; Critical = all outputs off, manual reset needed. Trigger

thresholds follow Table D and Table E. Latched protection = charging or discharging stopped, explicit reset needed after the cause is removed.

Table A – Error codes

Code	Name	Description	Severity
E01	OVERVOLTAGE	A cell voltage is above the maximum cell voltage. Charging stopped.	Protection
E02	UNDERVOLTAGE	A cell voltage is below the minimum cell voltage. Discharging stopped.	Protection
E03	OVERCURRENT_CHARGE	Charge current is above the charge current limit.	Latched protection
E04	OVERCURRENT_DISCHARGE	Discharge current is above the discharge current limit.	Latched protection
E05	OVERHEATING	Battery temperature > 55 °C. Battery cut off, fans 100 %.	Protection
E06	UNDERTEMPERATURE	Battery temperature too low for charging. Charging stopped.	Protection
E07	SHORT_CIRCUIT	Short circuit detected at the output. All outputs off.	Critical
E08	ESTOP	Emergency stop pressed. All outputs off.	Critical
E09	TEMP_SENSOR_FAULT	A temperature sensor is open, shorted or implausible.	Critical
E10	AFE_COMM_FAULT	MCU cannot communicate with the cell-monitoring chip (AFE).	Critical
E11	PI_COMM_LOST	No data exchange between MCU and Raspberry Pi.	Warning
E12	FAN_FAULT	Fan does not reach expected speed.	Warning
E13	CELL_IMBALANCE	Voltage difference between cells larger than allowed.	Warning
E14	WATCHDOG_RESET	MCU was restarted by the watchdog.	Warning
E15	CONFIG_INVALID	Battery configuration failed the start-up check. Outputs stay off.	Critical
E16	TERMINAL_OVERTEMP	Terminal joint (NTC2) too hot, or much hotter than the cells. All outputs off.	Critical
E17	ENCLOSURE_OVERTEMP	Air downstream of the battery (NTC3) too hot. Fans 100 %.	Warning
E18	CHARGE_OVERTEMP	Battery temperature too high for charging (> 50 °C). Charging stopped.	Protection
E19	DISCHARGE_UNDERTEMP	Battery temperature too low for discharging (< −10 °C). Discharging stopped.	Protection

Met opmerkingen [SR32]: Table A said Protection (auto-recovery) but Table E requires 'cause removed + explicit reset' for OC. New severity 'Latched protection' matches Table E.

Met opmerkingen [SR33]: Table E rows Charge OT and Discharge UT, and E3.16 (NTC2), had no error code.

3.5 Communication

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S5.1	The MCU shall send the measured voltage of each cell group to the Pi in every telemetry frame (S5.9).	–	Must	D	5.1
S5.2	The MCU shall send the battery-pack voltage to the Pi in every telemetry frame (S5.9).	–	Must	D	5.2

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S5.3	The MCU shall send the battery current to the Pi in every telemetry frame (S5.9), with the measured sign of E5.8 (negative = charging).	–	Must	D	5.3
S5.4	The MCU shall send all measured battery temperatures to the Pi in every telemetry frame (S5.9).	–	Must	D	5.4
S5.5	The MCU shall send each new or cleared fault to the Pi within the fault latency given in S5.8.	≤ 100 ms	Must	D	5.5
S5.6	The MCU shall send the system state, as one of the states in S8.9, in every telemetry frame.	–	Should	D	5.6
S5.7	MCU–AFE communication shall use I²C.	100 kbit/s Read every 100 ms	Must	I	FB
S5.8	MCU–Pi communication shall use UART (8N1). The MCU shall send a telemetry frame at the normal transmission rate, and shall send fault messages within the fault latency after detection when the link is available.	115200 baud Telemetry: every 1 s ± 10 % (TBC) Faults: ≤ 100 ms	Must	T	FB
S5.9	UART frames shall be binary: header + length + sequence + payload + CRC-16/CCITT-FALSE. The payload shall contain the 4 cell-group voltages, current, temperatures, state and fault flags. If a frame fails the CRC or length check, the Pi shall discard it and increment the communication error counter (S9.11).	CRC-16/CCITT-FALSE	Must	T	FB
S5.10	The UART protocol between C firmware and Rust application shall be defined in one document (/protocol): message IDs, field order, data types, units, byte order (little-endian) and checksum. Both sides shall be implemented from this document.	–	Must	I	8.12
S5.11	The Rust application shall have tests that decode reference frames produced by the C firmware, so a protocol mismatch between the two languages is caught before deployment.	–	Should	T	8.13

Met opmerkingen [SR34]: Sign follows V.2_claude E5.8 (negative = charging).

Met opmerkingen [SR35]: DECISION NEEDED: I²C or SPI. Rewrite once decided.

Met opmerkingen [SR36]: Open item 9: normal rate proposed as 1 s, the minimum that meets the 1 s update in Table B.

Met opmerkingen [SR37]: Behaviour for corrupt frames was missing; testable by sending bad frames.

3.6 Fan control

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S6.1	Fan control shall run on the Pi, connected to a fan controller.	–	Must	I	9.1
S6.2	The Pi shall set the fan duty from the configurable temperature/duty table (Table C), received from the MCU (S2.4), with linear interpolation between points.	≤ 8 points Duty accuracy ± 1 %	Must	T	9.2

Met opmerkingen [SR38]: Note: cooling now depends on the Pi software. See S3.8 for what happens when the Pi is lost.

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S6.3	Fan duty shall be the higher of two table values: one from the highest battery temperature (NTC1, Table C) and one from the air temperature downstream of the battery (NTC3).	max(NTC1 table, NTC3 table) NTC3 table: Table C2	Must	T	9.3
S6.4	While the fans are off, they shall switch on at 30 % duty when the temperature reaches the switch-on point. While the fans are on, they shall keep running (at least 30 %) until the temperature falls to the switch-off point. At or above the full-speed point, the duty shall be 100 %.	On: ≥ 30 °C Off: ≤ 27 °C 100 %: ≥ 45 °C	Must	T	9.4
S6.5	When E05 is active, the fans shall run at 100 % until the Table E recovery condition for discharge OT is met. An active E-stop overrides this: fans off (E7.7).	Cut-off: > 55 °C Release: ≤ 50 °C for 5 s	Must	T	9.5
S6.6	If the duty cycle is at or above the minimum and the measured speed stays below the set fraction of the expected speed for that duty for the set time, warning E12 (fan fault) shall be raised. The expected speed per duty shall be stored in the fan table.	Duty ≥ 30 % Speed < 70 % of the speed measured at that duty during commissioning Time: 5 s	Should	T	9.6
S6.7	A hardware fallback independent of the Pi shall drive the fans to 100 %: the 4-wire PWM fans run at full speed when the PWM signal is lost (M5.5), and a thermostat switch near the battery forces the fans to 100 % above its set point. The E-stop overrides the fallback (E7.7).	Thermostat: 50 °C (TBC)	Should	T	M5.5
S6.8	When E17 is active, the fans shall run at 100 % until the Table E recovery condition for enclosure air OT is met.	On: NTC3 > 45 °C Off: NTC3 ≤ 40 °C for 5 s	Should	T	E3.17

Met opmerkingen [SR39]: Aligned with V.2_claude: proposal for open item 1. NTC3 table values still TBD.

Met opmerkingen [SR40]: DECISION NEEDED: open item 1 (highest battery temperature vs enclosure NTC3).

Met opmerkingen [SR41]: Behaviour inside the 27–30 °C hysteresis band was not stated. Test with a rising and a falling ramp.

Met opmerkingen [SR42]: Aligned with V.2_claude: cut-off '> 55 °C'; E7.7 now says the E-stop switches the fans off, even during E05.

Met opmerkingen [SR43]: Release was '< 50 °C' here but '≤ 50 °C for 5 s' in Table E. Now refers to Table E.

Met opmerkingen [SR44]: 'Set speed' was undefined. Measure the duty→rpm curve to fill the fan table.

Table C – Fan speed table

Max. battery temperature	Fan duty	Remark
≤ 27 °C	0 % (off)	Switch-off point (hysteresis)
30 °C	30 %	Switch-on point
40 °C	65 %	Linear interpolation
≥ 45 °C	100 %	Full speed
> 55 °C	100 %	E05 overheating: battery cut off

Table C2 – Fan speed table, enclosure air (NTC3)

NTC3 air temperature	Fan duty	Remark
≤ 32 °C	0 % (off)	Switch-off point (hysteresis)
35 °C	30 %	Switch-on point
40 °C	65 %	Linear interpolation
≥ 45 °C	100 %	Full speed
> 45 °C	100 %	E17 enclosure warning (E3.17, S6.8)

3.7 PC connection

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S7.1	The MCU (TI MSPM0G3507; LP-MSPM0G3507 LaunchPad for development) shall be programmable and debuggable through SWD: on the LaunchPad with the on-board XDS110 probe over USB, and on the BMS PCB through a 10-pin Cortex Debug header (3.3 V) using the LaunchPad XDS110 or another Arm SWD probe, with Code Composer Studio or Keil.	10-pin, 1.27 mm	Must	D	10.1
S7.2	Debug builds of the firmware shall send log messages to the PC over the debug probe (SEGGER RTT or SWO) or the service UART (printf-style logging). Release builds shall not depend on a connected PC.	–	Should	D	10.2
S7.3	The MCU UART debug pins (TX, RX, GND) shall be available on a header so a USB-to-serial adapter can be connected for service.	3-pin, 2.54 mm 3.3 V	Should	I	10.3
S7.4	Within the set time after power-on, the Raspberry Pi touchscreen shall show the monitor application full-screen without a user login.	≤ 60 s (TBC)	Must	D	10.4
S7.5	The firmware shall be updatable through the SWD connector (S7.1). A firmware version change shall be logged as an event (S9.10).	–	Should	D	Review

Met opmerkingen [CL45]: The MSPM0G3507 is an Arm Cortex-M0+, which has no SWO trace output. Debug logs therefore need the service UART (S7.3), or SEGGER RTT if a J-Link probe is used.

Met opmerkingen [SR46]: 'Automatically runs' had no time limit. S9.1 merged into this.

Met opmerkingen [SR47]: New: no requirement covered how firmware is updated.

3.8 User interface

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S8.1	The BESS shall display the battery voltage.	–	Should	D	7.1
S8.2	The BESS shall display the battery current.	–	Should	D	7.2
S8.3	The BESS shall display the battery temperature.	–	Should	D	7.3
S8.4	The BESS shall display the voltage of each cell group.	–	Could	D	7.4
S8.5	The BESS shall display detected battery faults.	–	Must	D	7.5
S8.6	The BESS shall indicate whether the system is charging, discharging or idle.	–	Should	D	7.6

Met opmerkingen [SR48]: S8.1–S8.6 duplicate S8.7/Table B (and S8.3 is Should while Table B is Must). Consider deleting them or marking them as parent requirements of S8.7.

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S8.7	The dashboard shall show the values in Table B with the given unit, resolution and update rate. 'Today' means 00:00–24:00 local time.	–	Must	T	11.1
S8.8	The current shall be shown with a sign: positive = charging, negative = discharging; the measured value (negative = charging, E5.8) shall be inverted for display. Power shall be calculated as pack voltage × current.	–	Must	D	11.2
S8.9	The system state shall be shown as one of: IDLE, CHARGING, DISCHARGING, FAULT, E-STOP or OFFLINE. Each state shall be entered and left according to the state table in the software documentation (S1.13). When several states apply, the priority shall be E-STOP > FAULT > OFFLINE > CHARGING / DISCHARGING > IDLE.	–	Must	D	11.3
S8.10	Each active fault shall be shown with error code, name, plain-language description, severity and time of occurrence. A critical fault shall be shown as a red banner on every page until it is cleared.	–	Must	D	11.4
S8.11	Values shall be coloured yellow at the warning level and red at a protection limit (Table E).	Yellow: cell ≥ 3.50 V or ≤ 2.90 V Yellow: temp. ≥ 45 °C or ≤ 10 °C Yellow: SoC < 20 % or > 85 % (TBC)	Must	D	11.5
S8.12	If no new data has arrived from the BMS within the timeout, the dashboard shall show “BMS communication lost” and grey out all live values.	3 s	Must	T	11.6
S8.13	The dashboard shall estimate time to empty (when discharging) and time to full (when charging) from current and SoC. Time to empty = remaining Ah / |I|; time to full = (100 % − SoC) × CAP_AH × SoH / |I|. The estimate shall be shown as hh:mm and hidden when |I| is below the set level.	Hidden if |I| < 0.2 A (TBC)	Could	D	11.8

Met opmerkingen [SR49]: V.2_claude keeps both E5.8 (negative = charging) and S8.8 (positive = charging on screen); this states the inversion explicitly (open item 2, option b).

Met opmerkingen [SR50]: Entry/exit conditions were undefined (only CHARGING had one, in E5.8). The state table must be written in /docs.

Met opmerkingen [SR51]: Warning (yellow) levels were not defined. SoC levels follow the E1.7 window.

Table B – Information shown to the user

Information	Unit	Resolution	Update
State of charge (SoC)	%	1 %	1 s
State of health (SoH)	%	1 %	1 day
Pack voltage	V	0.01 V	1 s
Current (+ charge / − discharge)	A	0.01 A	1 s
Power	W	0.1 W	1 s
Energy charged / discharged (today, total)	Wh	0.1 Wh	1 s
Charge in / out (today, total)	Ah	0.01 Ah	1 s
Each sensor temperature + maximum	°C	0.1 °C	1 s

Information	Unit	Resolution	Update
Each cell-group voltage + min / max / ΔV	mV	1 mV	1 s
Fan duty and speed	% / rpm	1 % / 30 rpm (2 pulses/rev, 1 s window)	1 s
System state	–	–	1 s
Active faults (code, name, time)	–	–	≤ 1 s after occurrence
Equivalent full cycles	–	0.1	1 day
Time of last BMS update	hh:mm:ss	1 s	1 s

3.9 Graphs / HMI

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S9.1	The HMI shall be an application dashboard served by the Raspberry Pi. (Start on boot: see S7.4.)	–	Must	D	12.1
S9.2	The dashboard shall work without internet access (all scripts and fonts stored on the Pi).	–	Must	T	12.2
S9.3	The dashboard shall have the pages: Overview (live values), Cells, Graphs, Fault log and System info.	–	Must	I	12.3
S9.4	Overview: SoC and SoH as gauges, pack voltage, current, power, maximum temperature, system state and active faults.	Refresh: 1 s	Must	D	12.4
S9.5	Cells: bar chart of the 4 cell-group voltages with minimum, maximum and ΔV highlighted, plus a table with each cell-group voltage.	4 groups	Should	D	12.5
S9.6	Graphs: time graphs of fan speed, temperature (all sensors), SoC, SoH, voltage, current and power.	–	Must	D	12.6, Notes
S9.7	Each graph shall have selectable time ranges. Tapping or hovering on a point shall show the exact value and time.	1 h (1 s data) 24 h 7 days 30 days	Should	D	12.7
S9.8	Graphs of 10-minute and daily data shall show the min–max band together with the average line.	–	Could	D	12.8
S9.9	Fault log: a table with timestamp, error code, name, severity, measured value at the moment of the fault and the time it was cleared, filterable by severity and date and sortable by column.	–	Must	D	12.9
S9.10	Event log: a table of system events (start-up, shutdown, state changes, fault resets, firmware version changes).	–	Should	D	12.10
S9.11	System info: firmware and Pi application version, battery configuration (cell count, CAP_AH, limits), uptime, cycle count, database size and communication error count.	–	Should	D	12.11

Met opmerkingen [SR52]: Start-on-boot duplicated S7.4; merged there.

Met opmerkingen [SR53]: The Pi has a touchscreen (S7.4), where hovering is not possible.

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S9.12	The dashboard shall be read-only by default. Actions (fault reset, Pi shutdown) shall require a confirmation dialog and a PIN. The PIN shall be stored hashed, and after the set number of wrong attempts PIN entry shall be locked for the set time.	PIN: 4–6 digits 5 wrong tries → 5 min lock (TBC)	Should	D	12.12

Met opmerkingen [SR54]: PIN rules and the meaning of 'shutdown' were undefined.

3.10 Data storage

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S10.1	Data shall be stored on the Raspberry Pi in an SQLite database in WAL mode, so a power loss loses at most the last record being written.	–	Must	T	13.1
S10.2	The most recent 1-second telemetry shall be kept in RAM for the live graphs and is not written to disk.	Last 1 h	Should	I	13.2
S10.3	Periodically, one record shall be stored with the minimum, average and maximum of: pack voltage, current, power, each temperature, each cell-group voltage, SoC and fan speed, plus Ah and Wh charged and discharged in that interval.	Every 10 min	Must	T	13.3, Notes
S10.4	Every day, old 10-minute records shall be compacted into one daily record (min/avg/max per value, plus daily Ah, Wh and equivalent cycles).	At 00:00 local time Records > 30 days	Must	T	13.4
S10.5	Every fault and event shall be stored within the set time after it is received, with timestamp, error code, severity and measured values. The fault/event log shall not be compacted or deleted automatically.	≤ 1 s	Must	T	13.5
S10.6	When a fault occurs, the 1-second data around the fault shall be saved as a fault snapshot.	60 s before to 60 s after	Must	T	13.6
S10.7	Data shall be kept for at least the retention periods.	10-min: ≥ 30 days Daily: ≥ 2 years Faults/events: ≥ 2 years	Must	A	13.7
S10.8	The database shall stay below the size limit (estimate: ≈ 2 MB for 30 days of 10-minute data, < 1 MB per year of daily data). The Pi shall use a sufficiently large storage medium.	< 200 MB after 4 years Storage ≥ 16 GB	Must	A	13.8
S10.9	SoH shall be calculated as measured capacity / CAP_AH × 100 %, where measured capacity is the Ah counted between two rest (open-circuit voltage) points at least the set SoC difference apart, divided by that SoC difference. Each SoH value shall be stored so its trend can be shown in the SoH graph.	Once per day ΔSoC ≥ 50 % (TBC)	Must	T	13.9, Notes
S10.10	All timestamps shall be stored in UTC (ISO 8601) and shown in local time.	–	Must	T	13.10

Met opmerkingen [SR55]: Timestamps are stored in UTC (S10.10), so 'At 00:00' needed a time zone.

Met opmerkingen [SR56]: 'Immediately' replaced by a time limit.

Met opmerkingen [SR57]: 'Measured capacity' had no method. A full 100→0 % cycle rarely happens inside the 20–85 % window (E1.7), so a partial-cycle method is proposed.

Met opmerkingen [SR58]: Split: the NTP/RTC part is now S10.16 with a measurable accuracy.

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S10.11	The user shall be able to export any time range and set of values as CSV: UTF-8, comma-separated, a header row with units (e.g. voltage_V), ISO 8601 timestamps.	–	Must	D	13.11
S10.12	The complete database file shall be downloadable from the System info page as a backup, and copyable from the Pi over an Ethernet cable (scp) as a database backup.	–	Should	D	13.12
S10.13	The CSV export (S10.11) shall open in Microsoft Excel with one column per value and the units in the header row.	–	Could	D	13.13
S10.14	A copy of the database shall be made automatically to a second location (USB stick or a second file on the SD card of the Pi).	Daily Keep last 7	Should	T	13.14
S10.15	Data may be uploaded to a cloud service (e.g. Firebase). Uploads shall be queued and retried when offline, shall never block local logging, and credentials shall not be stored in the source code.	–	Could	T	13.15, Notes
S10.16	The Pi shall synchronise its clock with NTP when a network is available, and keep time with an RTC module without a network.	With NTP: error ≤ 2 s Without: drift ≤ 2 s/day (TBC)	Must	T	13.10
S10.17	After a Pi restart, logging shall resume within the set time, and the gap shall be recorded as an event.	≤ 60 s (TBC)	Should	T	Review

Met opmerkingen [SR59]: 'By cable' was undefined. Ethernet/scp is a proposal – confirm.

Met opmerkingen [SR60]: 'Should be convertible' could not be tested.

Met opmerkingen [SR61]: S10.16 split from S10.10; S10.17 is new (data gaps after a Pi restart were not covered).

3.11 Remote website

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S11.1	The data uploaded per S10.15 (10-minute records, daily records, fault/event log and fault snapshots) shall be viewable on a website for TE employees.	–	Could	D	13.16
S11.2	The website shall only be accessible after login, either with TE company single sign-on or with individual accounts created by an admin. The website shall not be publicly accessible.	–	Must*	T	13.17
S11.3	The website shall have two user roles: Viewer (read-only) and Admin (manages users and email recipients).	–	Should*	D	13.18
S11.4	The website shall show the latest values (SoC, SoH, pack voltage, current, power, max. temperature, system state) together with the time of the last received update.	–	Could	D	13.19
S11.5	The website shall show the same graphs as the local dashboard, with selectable time ranges.	24 h 7 days 30 days 1 year	Could	D	13.20
S11.6	The website shall show the fault/event log as a table that can be filtered by date and severity and exported to CSV.	–	Could	D	13.21

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S11.7	10-minute records and fault events shall be uploaded within the set times.	Records: ≤ 2 min Faults: ≤ 30 s	Should*	T	13.22
S11.8	All data shall be sent over an encrypted connection (HTTPS/TLS). The Pi shall log in with a device-specific key stored in a protected file on the Pi, not in the source code. The key can be revoked if the Pi is lost.	–	Must*	I	13.23
S11.9	The website shall be read-only towards the BESS: it shall not be possible to send commands, change settings or reset faults on the BESS from the website.	–	Must*	T	13.24
S11.10	Moved to P1.7 (project requirement).	–	–	I	13.25
S11.11	Each BESS shall have a unique device ID so that the website can show more than one system.	–	Could	I	13.26
S11.12	Each website page shall load quickly and be usable on a phone.	Load < 3 s (Lighthouse 'Slow 4G') Width ≥ 360 px	Could	T	13.27

Met opmerkingen [SR62]: Load time needs a defined network condition to be measurable.

Live error status

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S11.13	The website shall show every active error (code, name, description, severity, time it started) shortly after it occurs, and move it to the fault log shortly after it is cleared.	≤ 60 s	Could	T	13.28
S11.14	The website shall show an overall status indicator: green = OK, yellow = warning, red = critical, grey = offline.	–	Could	D	13.29
S11.15	The Pi shall send a heartbeat to the cloud. If none is received within the timeout, the website shall show “Offline” with the last-seen time. An offline system shall never be shown as “no errors”.	Heartbeat: 60 s Offline after: 5 min	Could	T	13.30
S11.16	Error codes and descriptions on the website shall be the same as in the error code table used by the firmware and the local dashboard (Table A).	–	Could	I	13.31
S11.17	For each error code, the website shall show how many times it occurred recently.	Last 30 days	Could	D	13.32

Email alerts

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S11.18	When a Critical error occurs, an email shall be sent to every recipient on the alert list. Each recipient can also choose to receive emails for Warnings.	–	Could	T	13.33
S11.19	The alert list (name, TE email address, severity level) shall be editable by an Admin on the website without a software change.	1–10 recipients	Could	D	13.34

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
S11.20	The email shall be delivered shortly after the error occurs.	≤ 5 min	Could	T	13.35
S11.21	The email subject shall be “[BESS <device ID>] <severity> <code> <name>”. The body shall contain the description, time (local), measured value, system state, SoC and a link to the website.	–	Could	I	13.36
S11.22	The same error code shall not trigger another email within the repeat window. If it occurs again in that period, the next email shall state how many times it occurred.	30 min	Could	T	13.37
S11.23	A follow-up email shall be sent when the error is cleared.	–	Could	T	13.38
S11.24	If the BESS has been offline for too long, the cloud service (not the Pi) shall send an “offline” email.	> 15 min	Could	T	13.39
S11.25	Emails shall be sent by the cloud service, not by the Pi. The mail service credentials shall not be stored in the source code.	–	Could	I	13.40
S11.26	Every email sent shall be logged with time, recipient and error code.	–	Could	I	13.41
S11.27	The safety of the BESS shall not depend on the website, the internet connection or email delivery. All protection stays on the BMS.	–	Must	A	13.42

Met opmerkingen [SR63]: Safety statement should not be 'Could'. Test: disconnect network and Pi; all protection must still trip.

4 Project Requirements

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
P1.1	The project shall deliver: schematic, PCB files, BOM, block diagrams, software documentation, drawings and test results.	–	Must	I	FB
P1.2	Safety documents shall be prepared: written work instructions and checklists (SOPs), a risk assessment and emergency procedures.	–	Must	I	FB
P1.3	Components shall be sourced from approved/preferred suppliers such as Mouser or Farnell, using these availability criteria: in stock at ≥ 2 suppliers or ≥ 100 pcs at Mouser or Farnell, lead time ≤ 2 weeks, and not marked end-of-life or not-recommended-for-new-designs.	–	Should	I	FB
P1.4	TE Connectivity products (connectors, relays/contactors, sensors, etc.) shall be used where technically suitable, without compromising the technical requirements.	–	Should	I	FB
P1.5	The report shall document the upgrade path to the 12 A design current (which parts limit the current today and what has to change) and, for each key component, the backup choice and its effect on the rest of the system.	–	Must	I	New

Nr.	Requirement	Value	MoSCoW	Ver.	V1 ref.
P1.6	Key design choices (e.g. conductor material, wire, fuses, fans, sensors) shall be documented in a trade-off table that lists the options and the reasons for the choice.	–	Should	I	New
P1.7	Before company data is placed on an external service (e.g. Firebase), the hosting choice shall be approved by TE IT.	–	Must*	I	13.25 (was S11.10)

5 Open Items

#	Item	Affects
1	Fan control input and thresholds. S6.3/S6.4 use the highest battery temperature with fans on at 30 °C and 100 % at 45 °C. The temperature research proposes the enclosure sensor (NTC3) with fans on at 35 °C and 100 % at 45 °C. 9.3 was marked "!!!!!!!!" in V1. V3 proposal: fan duty = higher of the NTC1 table and an NTC3 table (S6.3); NTC3 values TBD. V3.3: decided: fan duty = higher of Table C (NTC1) and Table C2 (NTC3).	S6.3, S6.4, Table C
2	Current sign convention. E5.8 treats negative current as charging; S8.8 shows positive current as charging. Agree on one convention (or state that the AFE value is inverted for display). V3.2: resolved in S8.8: measured value negative = charging (E5.8), inverted for display.	E5.8, S8.8
3	Balancing current. 15–50 Ω at ~3.3 V gives ≈ 0.07–0.22 A, not the ~0.3–1 A written in V1. V3.4: 50–100 mA, 33–68 Ω (E4.3).	E4.3
4	Loss-of-data timeout was written as "> 500–300 ms". V3 proposal: 500 ms (E3.10).	E3.10
5	Short-circuit current threshold. V3.4: ≈ 60 A (E3.7).	E3.7, Table E
6	C-rate versus temperature (1C–2C). V3.4: not needed: charging only from 10 °C at ≤ 0.83C (Table D).	Table D
7	E-stop reaction time (50–250 ms) and fan behaviour during an E-stop. V3 proposal: < 100 ms (E7.4); fans OFF on E-stop (E7.7).	E7.4, E7.7
8	USB output: type, voltage, current, power, protocol and protection. V3.4: 1 × USB-A (5 V) + 2 × USB-C with PD; total ≤ 45 W today, ≤ 95 W after upgrade (E6.1).	E6.1
9	Normal UART transmission rate (written as "data/s"). V3.2: telemetry every 1 s ± 10 % (S5.8, TBC).	S5.8
10	Remaining TBDs: cabinet inner dimensions (measure), fuse type (benchmark), USB PD controller and per-port power (electrical), Table G component data and total heat (after component selection), mains cord confirmation; software TBC values to be confirmed by test.	Various
11	Pack continuous discharge: Table D gives 9.6 A per cell (28.8 A pack); team decision is 9.6 A for the pack. Confirm from the cell datasheet. V3.3: confirmed, pack 28.8 A (3C); the battery is not the system limit.	Table D, Table F1
12	Peak duration t_peak at 12 A (cell datasheet pulse rating). V3.3: t_peak = 10 s (system choice).	Table F1, E3.18
13	Fuse types, ratings, breaking capacity and locations (fuse benchmark). V3.3: F1 15 A, F2 7.5 A (15 A upgrade), F3 removed; breaking capacity ≥ 1000 A; type from benchmark.	E8.5, E11.6
14	Inner dimensions and material of the existing cabinet (measure; metal → PE bonding).	M1.2, E11.3
15	Wire type, cross-section and bend radius after wire selection.	E10.1, E10.3, M3.1
16	Storage SoC and battery storage temperature > 3 months (cell datasheet).	M4.2, M4.5
17	Slew-rate limits and stuck-sensor time: proposed values to be confirmed by test.	S3.10, S3.12

#	Item	Affects
18	NTC3 fan table values and the E17 limit. V3.3: Table C2; E17 at > 45 °C.	S6.3, E3.17
19	Hardware fan thermostat set point.	S6.7
20	Component temperature ratings and heat dissipation.	Table G, M4.1, M5.3

6 Rationale for key decisions

This section gives the reasoning behind each requirement value or design choice that is not self-evident, so reviewers and future teams can see why it was chosen and what would have to change if the assumption changes.

6.1 Electrical

Nr.	Decision	Reasoning
E1.2, Table D	4S3P LiFePO$_4$ pack, 12.8 V, 9.6 Ah	4 cells in series give 12.8 V, which suits standard 12 V parts (PSUs, SSRs, fans). 3 in parallel give 9.6 Ah. LFP is thermally stable and does not readily release oxygen when abused, which makes it one of the safest lithium chemistries, and it has a long cycle life.
E1.7	SoC window 20–85 %	Avoiding the very full and very empty ends slows ageing and leaves room for SoC estimation error.
E1.8, Table F1	Design current 10 A continuous / 12 A peak; today ≈ 5 A	The present USB load needs about 5 A. Rating the power path for 10/12 A means an upgrade only needs a larger F2 fuse and new settings. The battery (28.8 A) is not the limiting part.
Table F1	Order of the current levels	Each level sits above the one before it including tolerances, so normal operation never trips a protection. The fast electronic trips act before the fuse; the fuse is the last line and protects the wiring if the electronics fail.
E2.1	Cell voltage accuracy ±10 mV	The LFP voltage curve is very flat, so small voltage errors give large SoC errors. ±10 mV is also well below the 30 mV balancing start level.
E2.3	Current range ±20 A, 10 mA resolution	The range covers every trip level (up to 13.65 A) with margin; 10 mA resolution keeps coulomb counting (SoC) accurate.
E2.4, M2.3	Three NTCs: cell, terminal joint, air	NTC1 protects the cells, NTC2 catches a loose or corroded joint (local hot spot), NTC3 shows whether the cooling works.
E2.6	Temperature range −20 to 100 °C	The NTC2 limit is 70 °C. The range must reach past it, otherwise a hot joint would be reported as a sensor fault instead of over-temperature.
E2.7	1 temperature sample per second	Cell temperature changes about 0.003 °C/s, so 1 s is far faster than needed while costing almost nothing.
Table E	Cell OV 3.60 V / UV 2.70 V with hysteresis	Inside the cell limits (3.65 V / 2.5 V) with margin for measurement error. The recovery levels (3.45 V / 2.90 V) are different from the trip levels so the output does not switch on and off repeatedly.
Table E	Temperature trips: charge 5–50 °C, discharge −10–55 °C	Inside the datasheet limits (charge 0–55 °C, discharge −20–60 °C) with 5 °C margin for sensor accuracy and sensor lag.
E3.6, E5.3	Charge trip 9 A, charger 8.0 A	Tolerance stack: the trip (9 A ±5 % = 8.55–9.45 A) stays below the battery maximum of 9.6 A, and the charger (8.0 A ±5 % = max 8.4 A) stays below the trip, so normal charging never trips.
E3.6	Discharge trip 6.5 A today / 13 A upgrade	Above the load plus tolerance, below the fuse rating.
E3.7	Short-circuit trip ≈ 60 A	Well above the highest normal current (13 A plus inrush) and well below the smallest realistic short-circuit current (≈ 150 A), so it never trips falsely and always catches a real short.
E3.10	Data timeout 500 ms	The AFE is read every 100 ms, so 500 ms is 5 missed readings: one glitch is ignored, a real loss is caught long before temperature or voltage can change much.
E3.12	Recovery needs 5 s of valid data and a reset	One good reading is not enough proof that the fault is gone; the explicit reset makes restarting a deliberate action.
E3.14	No averaging; highest reading counts	Averaging would hide a single hot spot.
E3.16	NTC2: > 70 °C or > 15 °C above NTC1	The absolute limit catches an overheating joint; the difference catches a bad joint early, even when everything is warm.

Nr.	Decision	Reasoning
E3.17	NTC3 warning at 45 °C	The Raspberry Pi is rated for 50 °C air; warning at 45 °C leaves 5 °C margin.
E3.18	Overload: t_peak = 10 s	Allows real short peaks (a device starting up, capacitor inrush) but stops a sustained overload before the wiring and fuse heat up.
E4.2	Balancing start 30 mV / stop 10 mV	The start level is well above the ±10 mV measurement accuracy, so noise does not trigger balancing; the gap between start and stop prevents switching on and off.
E4.3	Balancing 50–100 mA (33–68 Ω)	Enough to correct a 1 % imbalance (≈ 0.1 Ah) in about 2 h, with only 0.16–0.33 W of heat per resistor.
E6.1	USB-C with Power Delivery, total ≤ 45 W today / ≤ 95 W upgrade	PD allows fast charging of phones and laptops. The shared power limit keeps the battery current within the fuse and wire ratings (≈ 5 A today, ≈ 10 A after the upgrade).
E7.2, E7.3	Hardware E-stop, 2 NC channels, latching	Hardware works even if the firmware hangs. Normally-closed contacts mean a broken wire also stops the system (fail-safe). Two channels give redundancy; latching prevents restarting by itself.
E7.4	E-stop reaction < 100 ms	The safety relay and SSRs react within tens of milliseconds; 100 ms is short for personal safety and can be measured with an oscilloscope.
E7.5	Reset only after release + RESET; output sensing	Restarting must be a deliberate action. SSRs have no feedback contacts, so measuring the voltage after each switch is how we confirm they are really off.
E7.6	K1/K2 rated ≥ 15 A	Everything in the power path is rated at least as high as the largest fuse that can be fitted.
E7.7	Fans off on E-stop	Something caught in a fan is one of the E-stop hazards. Battery heat stops when the current stops, so cooling is not needed in that moment.
E8.2	Below 2.5 V per group: no discharge, charging allowed	Discharging further damages the cells; charging is the only way to recover them.
E8.5	F1 15 A, F2 7.5 A (15 A upgrade), no F3	F1 at the battery protects the whole path, including the charge wire; the charger limits its own current, so a separate charge fuse adds nothing. F2 is sized for today's load (≤ 75 % of rating) and swapped for the upgrade. DC rating ≥ 32 V because DC arcs do not go out by themselves; breaking capacity ≥ 1000 A because a short can reach ≈ 530 A.
E10.1	Power wire 2.5 mm^2, ≥ 105 °C	After derating for 45 °C cabinet air and two circuits per duct it still carries ≥ 15 A (the largest fuse). 105 °C insulation gives margin over the ≈ 60–70 °C the wire can reach at upgrade current.
E10.2	Separate ducts, twisted pairs	Distance and twisting reduce noise picked up from power wiring and switching.
E10.3	Flame-retardant insulation	Limits fire spread along cable bundles.
E10.4	Keyed power connectors	Connecting a battery in reverse causes immediate damage or fire.
E10.5–E10.7	Noise and switching tests	Prove that the wiring choices work in the real build.
E11.1	Fixed cord through a cable gland	Team choice: no loose connector to unplug, and the gland's strain relief stops the cord being pulled out of its terminals.
E11.3	Metal parts bonded to PE	If a mains wire touches the cabinet, the earth connection makes the building breaker trip instead of leaving the cabinet live.
E11.4	Separate mains duct, IPXXB with door open	230 V is the only shock hazard in the cabinet, so it is kept apart and finger-safe during maintenance.

Nr.	Decision	Reasoning
E11.6	Mains fuse F0 T2A	Normal input is ≈ 0.85 A; a 2 A fuse never blows in normal use. Time-lag survives the PSU switch-on inrush.
E11.7	All DC circuits SELV (≤ 60 V)	Voltages this low are safe to touch, so only the mains part needs shock protection.

6.2 Mechanical and thermal

Nr.	Decision	Reasoning
M1.3	IP20 (door closed), IPXXB (door open)	Indoor office use, no water. IP2X keeps fingers out of the fans and live parts. A higher IP rating would need filters and gaskets that cost airflow, money and maintenance.
M2.1	DIN rail, wire ducts, three zones	Standard industrial mounting: parts are easy to replace, and separating mains, power and signal improves safety and reduces noise.
M2.2	E-stop on the left side wall, ≥ 0.6 m, red on yellow	Reachable with the door open or closed; 0.6 m is the EN 60204-1 minimum height for controls; red on yellow (ISO 13850) is instantly recognisable.
M2.3	NTC1 on a middle cell, NTC3 downstream of the battery	A middle cell is the least cooled and hottest; NTC3 measures the air after it has passed the battery.
M2.4	Airflow: battery first, bottom in, top out	The battery gets the coolest air. Hot air rises, so the fans and natural convection push in the same direction.
M2.5–M2.9	Serviceability, labels, fixing, hinge loops, removable screen	Any part can be replaced without removing others; labels speed up fault finding; nothing loose or hanging on its wires; the previous team's screen was glued in and could not be removed without damage.
M3.1	Bend radius ≥ 4 × OD fixed, ≥ 10 × OD flexing	Tighter bends damage insulation and strands; cables at the door hinge flex every time it opens, so they need a larger radius.
M3.2	Wire duct fill ≤ 50 %	Leaves room for wires added during testing and keeps heat from building up in the duct.
M4.1	Room 0–35 °C, charging 10–35 °C	Charging stops at 50 °C battery temperature; the cabinet makes the battery up to 10 °C warmer than the room (M5.1); 5 °C margin gives 35 °C. Below 10 °C charging is blocked by the protection.
M4.2, M4.5	Storage > 3 months 0–25 °C, at 50 % SoC, checked every 3 months	Cells age slowest when stored cool and about half full. The datasheet gives no long-term value, so the team chose a conservative indoor range. LFP self-discharge is small, so a 3-month check is enough.
M4.3, M4.4	Humidity and altitude per IEC 61010-1	Standard "normal environmental conditions" for indoor lab equipment, so no extra testing is needed.
M5.1	NTC1 rise ≤ 10 K above inlet air	This number sets the 35 °C maximum room temperature, and it can be tested at room temperature without a climate chamber.
M5.2	≤ 5 °C between cells	Common battery thermal-management guideline: uneven temperatures age cells unevenly, which causes imbalance.
M5.3	≥ 5 K below component ratings	Datasheet ratings are limits, not targets.
M5.4, M5.5	Fan failure and fail-safe	Cooling must not depend on a single fan or on the Pi.
Table C2	NTC3 fan table: on at 35 °C, 100 % at 45 °C	From the temperature research. The cells have a large thermal mass and react slowly; the air reacts fast, so the fans respond to electronics heat before it soaks into the battery.
S6.7	Fan thermostat at 50 °C	It should only act if the Pi has failed: above the 45 °C full-speed point and below the 55 °C cut-off, allowing for the switch tolerance.

6.3 Software

Nr.	Decision	Reasoning
S1.6	No dynamic memory or recursion in firmware	Memory use is fixed and predictable, so the safety firmware cannot crash from running out of memory.
S2.2	Current limits as C-rates	Changing the battery only requires changing CAP_AH; no ampere values hidden in the code.
S3.8	Stop charging if the Pi is lost	Fan control runs on the Pi. Without it cooling is not guaranteed, and charging is the case with the most heat.
S3.10	Slew limits 1 / 3 / 2 °C per second	Far above how fast temperatures can really change (≈ 0.003 °C/s for a cell) and above reading noise (≈ 0.2 °C per sample), so only faulty readings are rejected.
S3.11	Median filter of 5 samples	Removes single spikes; the ≈ 2 s delay it adds is small compared to how slowly temperatures change.
S3.12	Stuck sensor: raw ADC unchanged for 10 min	A healthy sensor always shows a little noise in its raw value; the filtered value can legitimately stay constant in a steady room.
S3.13	5 rejected readings within 60 s → sensor fault	An occasional rejected reading is normal; repeated ones mean the sensor or its wiring is faulty.
S3.14	Watchdog 500 ms	A hung MCU is restarted quickly, and outputs stay off after the restart.
S5.7	MCU–AFE over I^2C	Team choice: fewer wires, and 100 kbit/s is more than enough to read the AFE every 100 ms.
S5.8	Telemetry every 1 s, faults within 100 ms	Matches the 1 s dashboard update; faults are sent immediately instead of waiting for the next frame.
S5.9	CRC on every frame	Detects corrupted messages so wrong values are never shown or stored.
S7.1	MCU TI MSPM0G3507, programmed over SWD	The LaunchPad's built-in XDS110 programmer can also program the custom BMS board, so no extra programmer is needed.
S8.12	"Communication lost" after 3 s	Three missed 1 s updates in a row means a real loss, not a single dropped frame.
S9.12	PIN for fault reset and shutdown	Prevents accidental or unauthorised resets from the touchscreen.
S10.1	SQLite in WAL mode	A power cut loses at most the record being written.
S11.27	Safety never depends on the website	Internet or cloud problems must never affect protection; all protection stays on the BMS.

6.4 Project

Nr.	Decision	Reasoning
P1.3	Supplier criteria	Parts must be buyable in time, have a second source, and not be about to be discontinued.
P1.4	TE products where suitable	Project goal of the sponsor, TE Connectivity, as long as the technical requirements are still met.
P1.6	Trade-off table for key choices	Shows why each choice was made, so TE and future teams can follow and reuse the reasoning.
P1.7	TE IT approval before external hosting	Company data on an external service needs approval from TE IT.